c)

$$E[W_t] = \underbrace{E[Z_t]}_{0}\,\underbrace{E[Z_{t-1}]}_{0} \quad \text{(linearity)}$$

$$E[W_t] = 0$$

$\forall\, k \in \mathbb{N}$

$$\text{cov}(W_t, W_{t-k}) = E[W_t W_{t-k}] - \underbrace{E[W_t]\,E[W_{t-k}]}_{0}$$

$$= E[Z_t Z_{t-1} Z_{t-k} Z_{t-k-1}]$$

$$= \begin{cases} \sigma^4, & \text{if } k = 0 \\ 0, & \text{else} \end{cases}$$

$\gamma_W(k)$ depends only on the lag k

conclusion: $\{W_t\}$ is stationary.

d)

$$E[R_t] = \cos(ct)\underbrace{E[Z_t]}_{0} + \sin(ct)\underbrace{E[Z_{t-1}]}_{0}$$

$$= 0$$